## EXPERIMENT 2

# Semantic Network Using NetworkX

## Aim

To represent concepts and their labeled relationships as a directed semantic network using Python and the NetworkX library.

## Software Required

Python 3.x and NetworkX. Install NetworkX using: pip install networkx

## Procedure / Algorithm

1. Import the NetworkX library.
2. Create a directed graph using nx.DiGraph().
3. Add the concepts and labeled relationships: Dog is-a Mammal, Mammal is-a Animal, and Dog has-instance Tommy.
4. Display each relationship by iterating through the graph edges and their relation attributes.
5. Run the program and verify the console output and semantic network diagram.

## Python Code

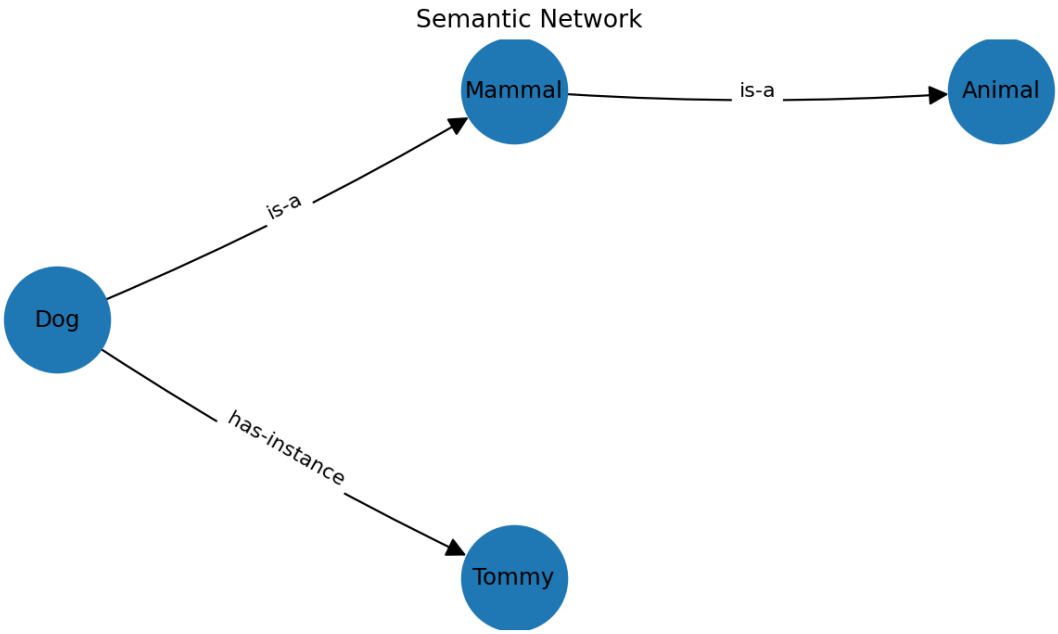
```
import networkx as nx

G = nx.DiGraph()
G.add_edge("Dog", "Mammal", relation="is-a")
G.add_edge("Mammal", "Animal", relation="is-a")
G.add_edge("Dog", "Tommy", relation="has-instance")

print("Semantic Network Relationships:")
for source, target, data in G.edges(data=True):
    print(source, "--", data["relation"], "-->", target)
```

## Output Screenshot

Figure 1: Directed semantic network showing concepts and labeled relationships.




## Console Output

Semantic Network Relationships:
Dog -- is-a --> Mammal
Dog -- has-instance --> Tommy
Mammal -- is-a --> Animal

## Result

The semantic network was successfully created using NetworkX. The directed graph represents Dog as a Mammal, Mammal as an Animal, and Dog as having the instance Tommy. The relationships were displayed with their labels.